



Immigrants and firms' outcomes: Evidence from France



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ABSTRACT

In this paper we analyze the impact of an increase in the local supply of immigrants on firms' outcomes, allowing for heterogeneous effects across firms. Using micro-level data on French manufacturing firms spanning the period 1995–2005, we show that a supply-driven increase in the share of foreign-born workers in a French *département* (a small geographic area) increased the total factor productivity of firms in that *département*. We also find this effect to be significantly stronger for firms with low initial productivity and small size. The positive productivity effect of immigrants was also associated with faster growth of capital, larger exports, and higher wages for natives. Highly-skilled natives, in competition with immigrants, moved towards firms hiring fewer immigrants spreading positive productivity effects to those firms too.

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1. Introduction

This paper estimates the effects of an increase in the supply of immigrant workers on firm-level productivity, investments, employment and exports. We also analyze whether these effects differ across firms depending on their initial productivity and size. To this end, we use French firm-level data for the period 1995–2005. Recent papers have found *indirect* evidence that immigrants benefit firms' productivity. Specialization of natives and immigrants in different and complementary tasks (Peri and Sparber, 2009) may increase production efficiency. Highly skilled foreign individuals contribute to innovation and productivity growth at the local level (Kerr and Lincoln, 2010; Peri et al., 2015). There is, however, scant *direct* evidence of the impact of immigrants on firms' outcomes because of limited data and identification issues at the firm level. When analyzing the impact of immigration at the firm level, one has to account for the heterogeneity that drives differences in firm-specific responses to a local change in immigrant supply. Simply analyzing the correlation between immigrant workers' employment and performance of firms is likely to be uninformative of the effect of the first on the second.

New data on French manufacturing firms and an identification strategy based on country-of-origin-driven variation in the supply of immigrants across geographic areas, that we call districts (*départements* in French),¹ allow us to focus on firm-specific effects of an increase in the local labor supply of immigrants. In particular, with our data, we can construct multiple outcomes at the firm level and follow them over time in a longitudinal dataset. Merging firm-level data with Labor Force Survey (LFS) data we can also construct an instrument for the local change in the supply of immigrants using a shift-share

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¹ France is administratively divided into 100 *départements* (including oversea territories).

method based on the pre-existing immigrants' location interacted with changes in the aggregate inflows of immigrants by country of origin.² We also control for several time-varying, firm-level characteristics, and for firm fixed effects to reduce the remaining concerns of endogeneity and omitted variable bias.

In our paper we analyze whether hiring of immigrants and other firm-level outcomes, in response to an increase in the local supply of immigrant labor, depend on initial productivity or size. If the hiring of immigrants is concentrated among firms that are initially more productive this may promote further agglomeration and increase the heterogeneity across firms. If, instead, it benefits less productive (through cost-cutting and complementarity), it may prevent job destruction and encourage growth of employment and output of less productive firms, reducing the heterogeneity across firms. In both cases it may also affect the wage of natives in those firms.

We first show that, based on aggregate statistics, the net immigrant inflow in France during the 1995–2005 period was skill-intensive (see Section 3.2). The share of immigrants with a high school diploma or less decreased from 74% in 1995 to 58% in 2005, while those with a 3-year university diploma grew from 9% to 14% of the foreign population.³ Moreover the employment of foreign-born workers increased significantly in occupations with high cognitive and analytical content relative to those with high manual content. Hence, the findings of this paper relate to the effects of high-skilled immigration.

We introduce two novelties relative to the existing literature. First, we focus on several firm-level outcomes such as productivity, investments and exports, which have been rarely considered in relation to immigration. Second, we allow different responses to a common increase in the local labor supply of immigrants, based on initial firm heterogeneity (in productivity and size). A supply-driven increase in immigrants in a local labor market increases the average employment of foreign-born in firms, as expected, but this effect turns out to be stronger for firms that initially had low productivity levels. We also show that firms with initially low productivity levels experience the largest positive productivity effects associated with the increase in immigrant supply. Other firms' outcomes related to productivity, such as domestic market share, exports, capital investments and probability of exiting the market are affected in similar ways by the increase in the supply of immigrants. These results are consistent with the idea that the productivity gains may be in part driven by skill-capital and skill-technology complementarity within the firm.

We find no evidence of a decline in native wages as a consequence of hiring immigrants. Rather, coherently with the higher productivity, the average wage of natives increases within a firm. Also the composition of native workers across firms changes as a consequence of immigration. Highly educated native workers are pushed into firms that hire fewer immigrants. Finally, new immigrants have a small but significant effect in reducing firm exit and their inflow increases the size of small firms relative to large firms. These differential effects and the reallocation of workers across firms imply a small average aggregated productivity effect in the local labor market, as immigration promotes the convergence of firms' productivity (and size) by stimulating small, initially less productive firms more than larger and initially more productive ones. If one were to analyze only local area effect, these rich inter-firm dynamics would be lost. We are not aware of any other study that has identified these effects.

The rest of the paper is organized as follows. Section 2 puts this paper in the context of the existing literature and introduces a simple framework to discuss the channels explored in the empirical analysis. Section 3 presents the data and stylized facts. Section 4 discusses the empirical specifications and the identification strategy and its limitations. Section 5 describes and comments on the estimates of the effects of immigration on firm outcomes. Section 6 concludes the paper.

2. Framework and literature review

While there is an abundance of studies on the effect of immigrants on the labor market outcomes of natives,⁴ much less is known about their firm-level effects. Recently some studies have emphasized mechanisms that would allow the absorption of immigrants within firms without a negative wage impact. Examples are the response in task specialization of native workers (Ottaviano et al., 2013; Peri and Sparber, 2009), the change in adopted production techniques (e.g. Lewis, 2011), and the growth in number of establishments (Olney, 2013). As a result, some recent studies have focused on firm-level evidence concerning the effect of immigration. Malchow-Møller et al. (2012), using data from Danish firms, find that immigrants substitute for natives within individual firms. Martins et al. (2012), using Portuguese data, find no effect of immigrants on the employment of natives at the firm level. These studies focus exclusively on the wage and employment effects of immigrants extending the classical labor market approach to firm-level data. Hatzi Georgiou and Lodefalk (2016) and Hiller (2013) consider another firm-level outcome, and analyze the effect of immigrants on firm-level exports. Ghosh et al. (2014) and Kirk et al. (2014) study the effect of a reduction in skilled immigrant visas on innovation, productivity, revenues and profits at the firm level. Ghosh et al. (2014) find a significant reduction in size and productivity for firms that heavily depend on foreign

² Our identification method is not new, as it builds on a traditional instrument proposed by Altonji and Card (1991) and then Card (2001). We perform several tests that show our instrument to be consistent with the exclusion restriction (i.e. be valid instrument).

³ From Labor Force Survey data we know that the share of immigrants in the group of tertiary educated increased from 5.7% to 7.1% over the period 1995–2005.

⁴ There are several studies analyzing the labor market effect of immigration at the national and local level. Seminal studies are Card (2001), Borjas (2003) and Ottaviano and Peri (2012). Recent surveys include Longhi et al. (2005) and Lewis and Peri (2015). Edo and Toubal (2015) is a recent paper using individual level data for France which finds that immigration decreases the wage of highly educated native workers by about 1% and contributes to an increase in the wage of low-skilled workers by 0.5%.

skilled workers following the decrease in visas.⁵ Kirk et al. (2014) do not find any impact on innovation at the firm level from hiring fewer skilled foreign visa holders. The policy context of our analysis differs from the US case because most of the increase in foreign-born workers in France during the 1995–2005 period was due to the free mobility within the newly constituted European Union. In the US, firms previously hiring workers were at a relative advantage for hiring more when the quota increased (or at disadvantage when it decreased). In France this is not the case, EU citizens were easily available to firms when barriers to mobility were removed in 1992.

Our analysis is focused on the impact of immigrants on firm-specific outcomes. We quantify the effect of an exogenous increase of immigrants in a local labor market on firms' productivity, and study whether this effect is heterogeneous across firms with different levels of initial productivity. To capture the variety of channels through which immigration may affect productivity and natives' wages in heterogeneous firms we consider the production function of a generic firm i within a labor market. We indicate with $s = \frac{F}{N+F}$ the share of foreign-born workers in that labor market, where N and F indicate the employment of natives and foreign-born, respectively. Firms can adjust their technology and efficiency responding to the native/immigrant composition of the labor force so that firm-specific productivity may depend on (s), via externalities. The presence of more immigrants may also increase the number of foreign workers hired by the firm (f_i), and it may also generate productivity gains (through specialization, complementarity and optimal task allocation) that may vary across firms. We allow for immigrants and natives to be imperfect substitutes in a CES production function (as in Ottaviano and Peri, 2012, or Docquier et al., 2014) and we allow firms to benefit from migrants' externalities within the region. We represent this using the following production function for firm i :

$$Y_i = A_i(s)(\beta f_i^{\frac{\sigma-1}{\sigma}} + (1-\beta)n_i^{\frac{\sigma-1}{\sigma}})^{\frac{\sigma}{\sigma-1}} \quad (1)$$

In Eq. (1), we are assuming that hired immigrants f_i and natives n_i are combined into output (Y_i) with an elasticity of substitution equal to $\sigma > 1$ so that the productivity of native workers may depend on immigrants through complementarities. Immigrants have a specific productivity β that may differ from that of natives ($1-\beta$). The productivity of the firm also depends, however, on the proportion of immigrants in the local labor market, s , through skill spillovers and externalities. While the CES component allows a positive wage and labor productivity effect of immigrants on natives within the firm through complementarity, the productivity spillover may extend to the whole area and it may vary across firms (hence the subscript i). Dividing Eq. (1) by $(n_i + f_i)$ implies that output per worker, y_i , and hence labor productivity of firm i , depends on the share of immigrants in the firm, s_i and in the region s as follows:

$$y_i = \frac{Y_i}{n_i + f_i} = A_i(s)(\beta s_i^{\frac{\sigma-1}{\sigma}} + (1-\beta)(1-s_i)^{\frac{\sigma-1}{\sigma}})^{\frac{\sigma}{\sigma-1}} \quad (2)$$

The first term, $A_i(s)$ captures the effect of the share of immigrants, through potential externalities, on the productivity of firm i . The intensity of this externality can be different across firms. The second term, in parentheses, captures the within-firm complementarity effect between immigrants and natives. It affects the average labor productivity of a firm and it has an internal maximum (between 0 and 1) which is a value of s_i corresponding to the “optimal” combination of immigrants and natives. This optimum is larger the larger the relative productivity of immigrants β . For equal productivity of immigrants and natives ($\beta = 0.5$), the second term is maximized at $s_i = 0.5$. Hence, for similar productivity of immigrants and natives, an increase in the share of immigrants in the firm will likely increase labor productivity in a firm through this channel at levels of s_i smaller than 0.5, which is the case for most of the considered firms. At higher values of s_i , the contribution of that term can become negligible or even negative.⁶

The average effect of immigrants on productivity per worker in a local labor market will be the weighted average productivity of each firm, y_i , with weights equal to the firms' shares of total employment in the local labor market (district) $x_i = \frac{n_i + f_i}{N+F}$. Hence:

$$y = \sum_{i \in \text{Firms}} x_i y_i = \sum_{i \in \text{Firms}} x_i(s) y_i(s) \quad (3)$$

The notation above emphasizes that not just the productivity of individual firms, y_i but their relative size, $x_i(s)$, can be affected by the percentage of immigrants in the local labor market. Both the productivity and the size of a firm may vary in response to changes in the local share of immigrants. Individual firms hiring more immigrants may benefit from their internal complementarity with natives, or from their externalities. The presence of immigrants can also induce firm creation and destruction. Hence Eq. (3) allows for possible effects of immigrants on firm size and on firm survival that may be heterogeneous across firms.

Our empirical analysis investigates, first, Eq. (2) at the firm level. In particular, we analyze how a change in the local share of immigrants over time affects the productivity of firms, allowing different effects along the initial productivity distribution. In particular, considering the variation over time of a linear version of Eq. (2) we can write:

$$\Delta y_i = y_i^a \Delta s + y_i^b \Delta s_i \quad (4)$$

⁵ Ghosh et al. (2014) focus on a sample of all large US firms.

⁶ The wage of native workers in firm i is given by the following expression $w_{ni} = A_i(s)(1-s_i)^{-\frac{1}{\sigma}}(\beta s_i^{\frac{\sigma-1}{\sigma}} + (1-\beta)(1-s_i)^{\frac{\sigma-1}{\sigma}})^{\frac{\sigma}{\sigma-1}}$. The extra-term $(1-s_i)^{-\frac{1}{\sigma}}$ shows how the complementarity between immigrant and native workers affects native wages positively and more strongly than average productivity in the firm.

In (4) the operator Δ captures the change over time. The term $y_i^a \Delta s$ captures the firm-specific change triggered by a change in the share of local immigrants, due to externalities, and the term $y_i^b \Delta s_i$ captures the change in productivity in a firm due to increased share of its own immigrant workforce, due to complementarity. Hence y_i^a and y_i^b are partial derivatives of firm productivity with respect to the immigrant externality and complementarity effect, respectively. We are interested in analyzing how the effects vary depending on firms' initial heterogeneity. What expression (4) also makes clear is that even firms not directly hiring immigrants may be affected by their inflow in the local economy as their productivity may depend on s through local externalities, as shown in the first term $y_i^a \Delta s$.

The estimated impact of immigrants on individual firm productivity is only part of the total effect that immigration can have on average local productivity. By changing the competition environment it may also affect the creation/destruction of firms, the selection of natives among them, and the relative firm size. Hence, decomposing the effect of a change in s on the average local area productivity expressed by (3), gives the following equation:

$$\Delta y = \sum_{i \in \Delta Firms} x_i^{new}(s) y_i^{new}(s) + \sum_{i \in Firms} \Delta x_i(s) * y_i(s) + \sum_{i \in Firms} x_i(s) * \Delta y_i(s) \quad (5)$$

The decomposition emphasizes that there are three potential channels affecting the local productivity. Beginning from the third term we will analyze how the change in immigrant share in a labor market area affects individual firm productivity and how this effect differs across firms (as illustrated by Eq. (4)); this is the focus of our firm-level analysis. Then, the first term of Eq. (5) captures the effect on creation and destruction of firms ($\Delta Firms$) due to an increase in s . Finally the second term in Eq. (5) captures the effect of immigration in reallocating workers across existing firms, and changing their relative size (Δx_i).

This framework is much richer than the one used traditionally to analyze the labor market effect of immigrants, which constrains many of the above effects to be zero and focuses on the aggregate impact on employment and wages. Here, we highlight the existence of several potential channels through which immigration can impact productivity and the local employment/wages within and across firms. We also extend the analysis (empirics) to other productivity related important firm outcomes (such as size, capital investment and exports).

3. Data and stylized facts

3.1. Sources and summary statistics

To investigate the effect of immigrants on firms' outcomes, using a unique firm identifier (SIREN), we combined three datasets containing detailed firm level information on employment structure, balance sheet, and export values of French firms. Then, to build our instrumental variable (see details in Section 4.1), we merge these datasets with French Labor Force Survey (LFS) data containing representative information on the population of immigrants across French geographical areas.

3.1.1. The employment structure of French firms: DADS data

The "DADS Poste" database, *Déclaration Annuelle des Données Sociales*, consists of administrative files based on mandatory reports on employees' earnings and characteristics. It contains yearly information about the structure of employment for each establishment in the private sector in France. Each single observation in this dataset corresponds to a unique worker-establishment-year combination.⁷ In terms of employee characteristics, we have information on the gender, age, place of birth (native vs. foreigners), total yearly gross and net earnings, number of hours worked in the year and worker's occupation.⁸ The establishment is identified with a unique establishment-code called (SIRET,) whose first nine digits represent the unique identifier for each firms (SIREN).⁹ Since other firm-level sources provide data at the firm (rather than establishment) level, we aggregate DADS Poste data at the level of firm by using the SIREN identifier. Individual employees are not linked across years, but firms are, hence we can construct the employment characteristics of each firm for each year. In particular, we construct the following variables: (i) the firm specific share of immigrants over total employment, (ii) the average hourly wage, and (iii) the total employment of native workers in the firm.

In terms of coverage, the DADS Poste covers all firms in the French private sector, including those operating in the manufacturing, service, and primary sector. For this reason we use DADS Poste also to obtain the total number of immigrant workers in each district-year. In the period 1995–2005 analyzed here, the DADS Poste counts 13,922,675 observations (establishment-year combinations). This number drops after the aggregation at the firm-year level and the merge with EAE data containing information on the balance sheet of French firms (and available only for a sub-set of manufacturing firms with more than 20 employees – see next section). In Section A1, we provide detailed information on the sample size, merging procedure and representativeness of the sample.

⁷ This database includes different units organized by year and region (12 years, 24 regions) which implies 288 separate databases with more than 50 million observations per year.

⁸ Unfortunately DADS Poste does not provide information on the educational attainment or labor market experience of workers.

⁹ DADS provides information also on the geographic location of the establishment.

Table 1
Summary statistics for firm-year observations.

Variable	Observations	Mean	Std dev.	Min	Max
Employment of natives	119,950	149.95	519.47	1	63,229
Wage of natives (per hour worked)	119,950	11.21	5.27	2.72	161.21
Per-capita value added	107,859	38.07	92.06	0	14,594
Total factor productivity OP	104,850	6.03	1.18	0.86	13.30
Total factor productivity LP	98,572	255.60	652.2	0.00	3760
Domestic market share	118,924	0.00	0.001	0.00	0.049
Capital	119,950	10,676	130,147	0	1.3*10 ⁷
Number of plants	119,950	1.72	4.21	1	236
Total export value	119,950	6.96*10 ⁷	5.19*10 ⁸	0	8.55*10 ⁹
Export value per destination	87,701	4.13*10 ⁵	1.66*10 ⁶	0	9.57*10 ⁷
Number of export varieties	119,661	35.62	110.05	0	5183
Number of export markets	119,661	10.21	15.61	0	174
Share of foreign-born workers	119,950	0.05	0.09	0	0.99

Note: Individual observations are firms by year. The sample includes all firms in France in the period 1996–2005. The sources of the data are described in the text [Section 3](#) and in online appendix Table A.1.

3.1.2. Firms outcomes: balance sheet and international trade data

The other firms' outcomes analyzed in this paper come from two additional data sources covering the manufacturing sector in France. Trade variables are from the French Customs dataset,¹⁰ which provides records of all the French firms exporting goods during the 1995–2005 period. Exports – quantified in tons and valued in Euros (free-on-board) – are available by firm, year, product, and the country of destination.¹¹ A threshold of 1000 euros applies to any trade with extra-EU destinations (for European countries such threshold is higher and around 150,000 euros).¹²

The rest of the firm outcomes come from annual business survey EAE, *Enquête Annuelle d'Entreprise*. It provides balance-sheet data information (e.g. value-added, total sales, value of the assets included in the capital stock, use of intermediate goods) needed to compute all the remaining firm outcomes used in this paper: (i) domestic market share, (ii) value-added per worker, (iii) physical capital and (iv) total factor productivity. The domestic market share is calculated as the ratio between the domestic sales of a firm, and the total domestic sales of French firms.¹³ Capital is approximated by the value of tangible assets at the beginning of the year, and the value-added per worker is calculated as the total value-added of the firm (before taxes) divided by the total number of employees.¹⁴ We use the Olley–Pakes method to compute the TFP of firms.¹⁵ EAE data are available for a set of surveyed manufacturing firms with more than 20 employees over the period 1995–2005.

3.1.3. Matching firm-level datasets.

We match all these sources together by using the SIREN identification number. We therefore obtain an unbalanced panel dataset of firms over the 1995–2005 period containing information on the firms' characteristics and outcomes.¹⁶ However, some of the variables needed to compute the TFP measure (namely capital and investment) have lots of missing observations in 1995; for this reason our estimation sample in [Tables 5–12](#) (and A3–A11) is restricted to the period 1996–2005. In the appendix [Section A1](#) we provide further details on data coverage. The summary statistics of the variables used in the paper are reported in [Table 1](#).

3.1.4. Aggregated and Labor Force Survey data

Aggregate data on immigrant and native population used to construct the local supply of immigrant labor are obtained in each district-year from the aggregation of the comprehensive DADS *Poste* individual data. Notice that such aggregation is performed on the original DADS *Poste* data, before any merge with other firm-level dataset, to assure the widest coverage

¹⁰ Provided by the DGDDI (*Direction Générale Des Douanes et Droits Indirects*).

¹¹ Products are identified by an 8 digit code corresponding to the NC8 nomenclature. For the number of products exported one needs to account for the fact that the product nomenclature changes over time during the period considered. To avoid the problem of mistaking a change in code for a change in product we harmonized the product nomenclature in the dataset, expressing all export and imports at the HS6 1996 revision.

¹² This difference in declaration thresholds is not problematic in our setting since we use French Customs data to compute the total export values of firms across all products and destinations.

¹³ Domestic sales are total sales (from EAE) minus the total export value of the firm (from French Custom data).

¹⁴ The total value-added corresponds to the total value of production of the firm in a year net of total production cost (i.e. materials, intermediate inputs, capital, labor and other goods and services needed to produce).

¹⁵ We dedicate appendix [Section A2](#) to a detailed description of how we computed TFP of firms.

¹⁶ A standard concern in firm-level (unbalanced) panels is that we observe only surviving firms. In our framework, the arrival of new immigrants in the district might change the level of competition among firms in the same area, implying a survivor bias in our estimation. To check for this potential bias, we re-estimate all our regressions on a balanced dataset including firms which are present over the entire period. Results do not change. As a sample of this robustness check, in [Table A6](#) we show results on TFP using a balanced panel of firms.

Table 2

Share of total hours worked by foreign-born individuals in French manufacturing sector.

Occupation type	Share of hours worked by foreign-born		
	1995	2005	Change 1995–2005
Skilled (top 5 cognitive/manual occupation)	0.046	0.165	0.119
Unskilled (bottom 5 cognitive/manual occupation)	0.093	0.119	0.016
Skilled (top 10 cognitive/manual occupation)	0.038	0.135	0.097
Unskilled (bottom 10 cognitive/manual occupation)	0.089	0.115	0.026
Non-production workers	0.038	0.131	0.093
Production workers	0.093	0.118	0.025

Note: Authors' calculation on the DADS dataset. Production workers are those in occupation groups CS 62, 63, 64, 65, 67, 68 and 69 (namely skilled manual workers, handlers, storage and transport workers, agricultural workers, industrial skilled workers, motor-vehicle drivers, unskilled manual workers, and unskilled industrial workers). Other occupation groups are non-production workers. The top 10 occupation by cognitive/manual complexity are as follows: government officials, engineers and technical managers, business and administration professionals, commercial managers, professors and scientist professions, administrative workers in private sector, teachers, information professions, administrative workers, and technicians. The bottom 10 occupation by cognitive/manual complexity are the following: police and military, personal services workers, skilled manual workers, skilled handlers, storage and transport workers, foremen, agricultural workers, skilled industrial workers, motor-vehicle drivers, unskilled manual workers, and unskilled industrial workers.

of the immigrant and native population.¹⁷ The share of immigrants by district and origin country in 1990 (used to build our instrument) is based on Labor Force Survey (LFS) data – *Enquête Emploi*. LFS provides information on gender, employment status, occupation, age, and nationality of individuals in France. Although LFS contains information on both the nationality and the country of birth of the individuals, in this paper we use nationality to characterize the migrant status of individuals. We do so in order to be coherent with the definition of immigrants used in the DADS data.¹⁸ LFS data are based on less frequent population censuses – *Recensements de la population* – conducted in the years 1982, 1990, 1999 and 2004.¹⁹ Then, yearly updates between consecutive censuses are based on yearly micro-censuses.²⁰ To build our instrumental variable, we use the spatial distribution of immigrants across districts by nationality in the year of the macro population census of 1990 (we count only individuals between 16 and 64 years old). LFS data over the period 1995–2005 are used to compute the total number of immigrants in France by nationality in a given year.

3.2. Immigrants skills and firm heterogeneity

During the 1995–2005 period, the net inflow of immigrants in France was more concentrated among highly skilled, and unevenly distributed across districts. We do not have direct information on the education of workers in the DADS data, so in order to characterize the skill content of French immigration in the period 1995–2005 we rely on the complexity and communication-intensity of the tasks performed by workers (we follow [Peri and Sparber, 2009](#) and focus on the dichotomy between communication and manual tasks to classify jobs along the communication-intensity spectrum).²¹ Table 2 shows that immigrant inflows between 1995 and 2005 were much larger among the high-complexity than the low-complexity occupations. The table shows that the top 10 occupations by complexity index increased the percentage of immigrants in total hours worked by 9.7 percentage points (p.p.), while the share of hours worked by immigrants among low-complexity occupations increased only by 2.6 p.p.²² Similarly, distinguishing production (blue-collar) and non-production (white-collar) manufacturing workers, the first group's share of immigrants increased by only 2.5 p.p. during the 1995–2005 period, while the second group increased by 9.3 p.p. Fig. 1 shows the percentage composition of immigrants in France (from the IAB dataset) from 1980 to 2010 by schooling degree. The largest group is primary-educated immigrants, but their share has been declining fast, especially over the past 15 years. To the contrary, the immigration of college-educated people has been increasing rapidly (the top share in the bar chart). Finally, over the 1995–2005 period, the share of immigrants with a high school diploma (or lower) decreased from 74% in 1995 to 58% in 2005. The share of immigrants with at least a three-year university diploma went from 9% in 1995 to 14% in 2005.²³ Using Labor Force Survey data we also compute the share of immigrants over total population by education attainment. In the group of tertiary educated population, the share of

¹⁷ For a proper geographic (district) allocation of workers, the aggregation has been computed on the establishment level DADS *Poste* – before the aggregation at the firm level. This solves any issue related with multi-establishment firms having plants in different districts.

¹⁸ LFS identifies 27 non-French nationalities.

¹⁹ Each population census has a sampling rate of approximately 1/20.

²⁰ Each micro-census surveys approximately 100,000 housings that represent a sampling rate of 1/300.

²¹ To do so we combine measures from the O*NET database that associate the intensity of several skills to each occupation in the Standard Occupational Classification (SOC) – cross-walking this classification with the occupational structure provided by the DADS. To this end, we merge both classifications into the 1988 International Standard Classification of Occupations, ISCO-88.

²² The evidence is similar if we take the top-5 occupations by complexity index.

²³ The education composition of immigrants is also displayed in Table A2 of the appendix.

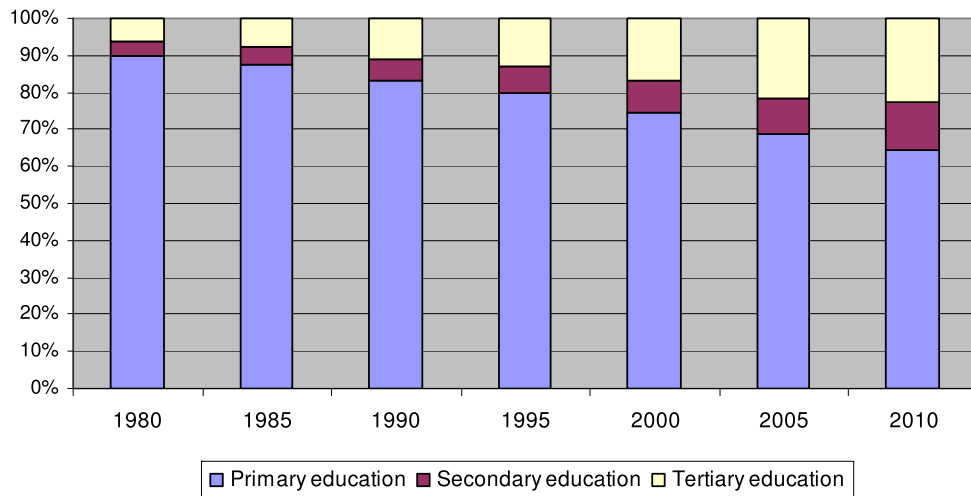


Fig. 1. Foreign-born in France, percentage of total by education category (primary, secondary and tertiary) Source: IAB Brain Drain Dataset.

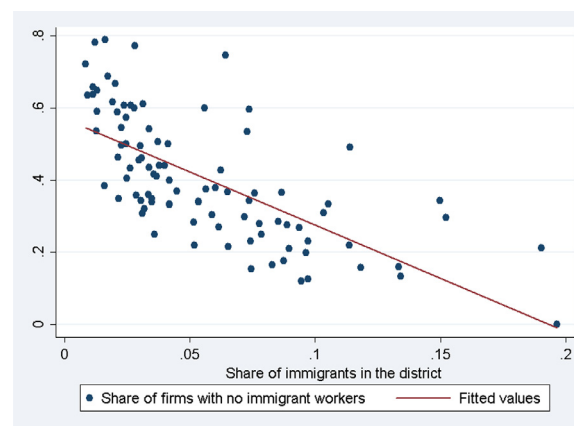


Fig. 2. Correlation between the share of firms with no immigrants and the immigrants' share in French districts in 1996. *Note:* Each point in the graph represents one district. The horizontal axis represents the share of immigrants in the population of the district while the vertical axis represents the share of firms with zero foreign employment among all firms.

immigrants increased from 5.7% to 7.1% over the period 1995–2005. In contrast, the share of immigrants in the group of unskilled population (with less than 3-year university diploma) decreased by 1 percentage point (from 6.7% to 5.6%). The net flow of immigrants into French employment during the period 1995–2005 period was skill-intensive.

A second typical feature of the inflow of immigrants in the period 1995–2005 was their uneven cross-regional distribution. Some regions (such as Champagne-Ardenne and Lorraine) maintained a low share of immigrant employment over the entire period. Other regions experienced increasing exposure to immigrants over time. As an example, the population percentage of immigrants in the Nord-Pas-de-Calais region increased from 6.3 points in 1995 to 16 points in 2005 (with an average yearly growth rate of 25%). Other regions simply increased the already-high share of immigrants over the same period. For example, in 1995 the Ile-de-France and Rhone-Alpes regions already had relatively large immigrant shares of 10% and 9.6%, respectively; that increased significantly to 21% and 18% in 2005, respectively.²⁴ The data also show a very large variation in the presence of foreign-born workers across firms. In 1996,²⁵ 37% of all French firms did not employ any immigrant workers (in 2005 this percentage was 40%). Such a share varied substantially across districts. In Seine Saint-Denis and Val d'Oise only 12% of all firms did not employ immigrants, while in Nierre and Haute Marne 78% of firms did not. Only in the South-East district of Haute-Garonne all firms employed at least one immigrant worker. The share of firms that do not hire immigrants is negatively correlated with the share of immigrants in the district – see Fig. 2. Also within districts there is a huge variation in the employment of immigrants across firms. The share of immigrant employment across firms is correlated to other firm characteristics. Table 3 shows the linear correlation coefficients across French manufactur-

²⁴ Data on the share of foreign population in 1995 and 2005 for all regions are shown in Table A1 of the online appendix.

²⁵ The year 1996 is the initial year for our estimation sample.

Table 3
Correlation between immigrants' share and firms' outcomes in 1996.

Variable	(1)	(2)
Total factor productivity OP	0.069*** (0.015)	−0.113*** (0.003)
Domestic market share	0.193* (0.103)	−0.667*** (0.015)
Employment	0.178** (0.086)	−0.623*** (0.018)
Capital	0.524*** (0.073)	−0.166*** (0.018)
Total export value	0.588*** (0.236)	−1.295*** (0.056)
Per capita revenues	0.152*** (0.054)	−0.060*** (0.012)
Explanatory variable	Immigrant share in the firm	Dummy for zero immigrants in the firm
Additional covariates	Region and sector fixed effects	

Note: Individual observations are manufacturing firms in France in 1996. The sources of the data are several and are described in the text [Section 3](#) and online Appendix A.1.

ing firms between immigrants and productivity proxies in year 1996.²⁶ The share of immigrants in the firm is significantly correlated with all the firm outcomes considered. Such correlation underscores the importance of controlling for firm characteristics and also the need for a sharper identification strategy that allows for heterogeneous response of firms and uses supply-driven inflow of immigrants in local labor markets.²⁷

4. Empirical framework

Our empirical model analyzes the response of firm outcomes to local changes in immigrant supply. We allow firms with low initial productivity (or size) to react differently to a change in the migrant labor supply within the district. To keep things simple we partition firms according to their initial productivity (or size) in 1996, and we allow for an additional effect of immigrants for firms below the median of initial productivity (or size).²⁸ We also consider specifically firms with initially zero immigrants. As shown in [Table 3](#), they are among the initially less productive firms. If immigrants have a productive complementary with natives, as hypothesized in the model above, then firms with initially zero immigrants may exhibit a particularly strong positive effect on average productivity when they move to hiring some immigrants.

Area-level changes in immigrant share, appropriately instrumented, will represent the exogenous supply shock for the firm. However, the size of the change in the firm's employment share of immigrants depends also on its own characteristics. Variations in the firm specific share of immigrants (within the district) are consistent with heterogeneous responses to common immigrant supply shocks in the district.²⁹ Thus, our baseline empirical model is the following:

$$(y)_{i,s,d,t} = \phi_i + \phi_{s,T} + \phi_{r,T} + \beta_1(s_{d,t}^{IMMI}) + \beta_2(s_{d,t}^{IMMI}) * I_i(k_{i,1996} \leq k) + \beta_3 X_{i,t} + \varepsilon_{i,t} \quad (6)$$

The variable $(y)_{i,s,d,t}$ represents an outcome relative to firm i in sector s , district d and year t (with $t = 1996\text{--}2005$). The term ϕ_i indicates a set of firm fixed effects, meant to capture the unobserved time-invariant firm-specific factors affecting the outcomes of the firms. Since, in our sample, every firm belongs to one specific sector and district (none of them move or changes sector), firm fixed effects controls also for district and sector-specific unobserved factors.³⁰ The “within” nature

²⁶ Each row of the table uses the variable listed in the first column as dependent in a regression, on region and sector fixed effects and on either the immigrant share in the firm employment (column 1), or on a dummy equal to one for having no immigrants and zero otherwise (column 2). The dependent variables are TFP, domestic market share, capital per worker and exports (all in logs). The first two are measures of labor productivity, while the other two variables are positively correlated with it.

²⁷ These are just correlations for descriptive purposes (no causal effect is claimed). Therefore, we will not use any of this cross-sectional association in our identification strategy.

²⁸ We will also check with a finer partition of firms by quartiles of initial productivity to show that the effect of immigration is actually monotonic in initial productivity. See [Table A7](#) in the appendix

²⁹ The few previous studies on the impact of immigrants at the firm level (e.g. [Malchow-Møller et al., 2012](#); [Martins et al., 2012](#)) combined the cross-sectional variation due to firm-level hiring and the variation of immigrant supply across localities. This may confound variation in local supply with variation in firms' response to it. In our approach we will consider only exogenous area-level variation of immigrants as a supply shock. Additionally, we will use the firm-level variation within a district to understand differential firm response. The trade literature has previously related export shocks to firm-level outcomes (such as productivity and wages). In that context (e.g. [Hummels et al., 2014](#)), it makes sense to consider a firm-specific export (or import) shock because of the differential exposure of each firm to the growth of specific foreign markets. In our case, immigration shocks are more reasonably considered as changes in the local availability of immigrant workers and, as such, they are common to all establishments in a labor market.

³⁰ In our sample each firm belongs to a specific district-sector over the entire period. For multi-plant firms operating in several sectors and districts we keep the sector and the district of the biggest plant (in terms of total number of workers). This is directly done by INSEE in providing the sector of each firm in the EAE data. We follow the same strategy in attaching a unique district to each firm. This approximation is not particularly problematic in our case

of this estimate, as we include firm effects, is consistent with Eq. (4) that describes how changes in firm's productivity depend on changes in local immigrant supply. The terms $\phi_{s,T}$ and $\phi_{r,T}$ are, respectively, sector-by-period and region-by-period fixed effects (one region is the union of 4–5 districts) and capture region- and sector-specific (time-variant) variables at a frequency of T years where $T = 2$.³¹ As one period lasts two years, those fixed effects fully absorb changes in French labor market legislation and/or sector technologies that evolve slowly over time.

The term $X_{i,t}$ captures a set of firm-specific time-varying controls, including a headquarter dummy (being one if the firm is the headquarter of a group), a foreign affiliate dummy (being one if the firm is an affiliate of a foreign multinational firm) and the firm's age. The explanatory variables of interest are: (i) $s_{d,t}^{IMMI}$, the share of immigrant population at time t in the district d ;³² and (ii) the interaction of this variable with the dummy $I_i(k_i, 1996 \leq k)$ that selects firms whose initial characteristic $k_i, 1996$ was below a certain threshold k usually chosen to be the median of that characteristic over the sample. This variable aims at capturing the heterogeneous effect depending on initial firm characteristics.³³

We estimate three different specifications depending on the choice of the initial characteristic k . First we consider the share of immigrants in firm i as of 1996, and in this case the threshold is zero; then the TFP of firm i as of 1996 and in this case the threshold is the sample median of the TFP; finally we consider the total employment of firm i in 1996 and also in this case the threshold is the sample median.

We analyze several outcomes, $(y)_{i,t}$, at the firm level. They are the total factor productivity (TFP), the domestic market share (as further proxy of productivity), the capital stock, the exit probability of firms, the number of plants per firm, the total exports, the number of foreign markets, the average wage of natives in the firms, and the employment of native workers in the firm. All those variables (except exit probability) are in logarithm and are positively correlated with productivity. Hence estimating a positive β_1 implies a positive effect of immigrant supply on these productivity outcomes, for firms with $k_i, 1996$ above the threshold. On the other hand a positive β_2 coefficient implies a stronger effect of immigrant supply for that outcome, in firms below the threshold. We estimate specification (6) on the full sample of firms. The standard errors are clustered at the district level to account for the correlation across firms and over time within a geographic area.

4.1. Identification issues and instrumental variable

As we include firm fixed effects in Eq. (6) and as the main explanatory variable varies at the district level, reverse causality concerns are small. Productivity changes of an individual firm do not have a significant effect on district-specific labor demand.³⁴ To further reduce omitted variable concern we also include fixed effects for region-period and sector-period unobservable variables. However, a remaining concern might come from the possibility that some non observable district-specific, time-varying factors may affect the changes in the demand of immigrants and the changes in productivity/labor demand of firms. They may generate a correlation between the residual $\varepsilon_{i,t}$, in Eq. (6) and the explanatory variable, $s_{d,t}^{IMMI}$. We address the remaining concern of endogeneity by using an IV strategy.

To do so, we use a shift-share instrument based on the pre-1992 distribution of immigrants. Pioneered by Altonji and Card (1991) and used in several studies since then (such as Card, 2001; 2009; Peri and Sparber, 2009, and Lewis, 2013, among others), this approach is based on the fact that, because of preferences, new immigrants tend to move where previous immigrants from the same country already live. This is because they learn of opportunities in those locations and because they enjoy the amenities of being near their co-nationals. Building on this idea we construct the imputed number of immigrants in district d , for each year $t = 1996 - 2005$, $\widehat{IMMI}_{d,t}$ as follows.

We allocate the total number of immigrants in France at time t from a given origin o , $\widehat{IMMI}_{o,t}$, in proportion to the share of that national group in the district as of 1990.³⁵ That year was before the creation of a common EU labor market in 1992 and hence pre-dates the large flow of EU immigrants. LFS data in 1990 correspond to the census data, and surveyed an extensive share of the total migrant population residing in France.³⁶ By summing across countries of origin o we obtain an imputed number of immigrants in each district:

$$\widehat{IMMI}_{d,t} = \sum_o \frac{IMMI_{d,o,1990,LFS}}{IMMI_{France,o,1990,LFS}} * \widehat{IMMI}_{o,t}. \quad (7)$$

because: (i) multi-plant firms operating in different district are only the 27% of total firms; (ii) the employment of the biggest plant (within a firm) is on average 26.9 times bigger than employment in other plants. In a robustness check in Table A8 we replicate the baseline regressions on a sub-sample of firms whose plants belong to a unique district, and results do not change.

³¹ Two-year period fixed effects imply that we have a dummy variable for a given region (or sector) every two years. We run several robustness checks by changing the duration of the period. See appendix Table A5 for TFP estimation results using, respectively, three- and five-year period definitions for the fixed effects.

³² We consider 92 non-overseas districts of France for which we have data on the 1990 stock of migrants by the LFS (needed to compute our instrumental variable). Thus, the explanatory variable of interest takes 920 different values (92 districts over ten years).

³³ This implies that results would be restricted to firms that existed in 1996. To attenuate this limitation, we run two robustness checks. First, we take the sub-period 2000–2005 and use the 2000 as starting year (using firms existing in 2000). Second, we run Eq. (6) on a balanced panel with firms existing over the entire period 1996–2005 (no entry/exit). Results for both robustness checks are reported in the online appendix Table A6 and confirm our baseline estimation results.

³⁴ In 1996 the average number of firms per district is 169, and each firm accounts (on average) for the 2.1% of the total workforce in the district.

³⁵ As a robustness check we alternatively use the spatial distribution of immigrants in 1982. See Table A9.

³⁶ Sampling rate of 1/20.

For the validity of this variable as IV, the size of existing immigrant communities in 1990 must not be correlated with district-specific labor demand changes between 1996 and 2005 (conditional on controls and fixed effects included in the Eq. (6)). As long as district-specific determinants of agglomeration of immigrant communities in 1990 are not strongly correlated with demand changes in the period 1996–2005, our IV can be considered valid. Moreover the aggregate inflow of immigrants from each origin country must be independent of the labor demand of one specific district. The validity of these assumptions are discussed in details and tested in the next section. Since the *total* number of immigrants in France by origin country and year, $\widetilde{IMMI}_{o,t}$, is not directly available for the period 1996–2005, we use survey LFS data to calculate $\widetilde{IMMI}_{o,t}$, as follows:

$$\widetilde{IMMI}_{o,t} = \frac{IMMI_{France,o,t,LFS}}{IMMI_{France,t,LFS}} * IMMI_{France,t,DADS}. \quad (8)$$

the first term in Eq. (8) is the share of immigrants in France from a given origin o from LFS in the year t ; while the second term in Eq. (8), $IMMI_{France,t,DADS}$, is the total number of foreign-born individuals working in France at time t in the manufacturing plus service sectors and is obtained from DADS data. This imputation allows us to use a more precise measure of total immigrants employed each year (from the DADS data) taking only the share by country of origin from the less precise LFS (yearly updates of LFS in the years between population census base on micro census. See Section 3.1.4).³⁷ We then impute the share of immigrants in each (district) as follows:

$$s_{d,t}^{IMMI} = \frac{IMMI_{d,t}}{IMMI_{d,t} + Natives_{d,1990}} \quad (9)$$

In expression (9) the native population is fixed at year 1990 so as to avoid spurious effects due to the potentially endogenous native population growth in the district over the 1996–2005 period. Internal mobility of native workers due to local demand shocks could introduce significant spurious bias in the instrument if one uses the contemporary (rather than initial) number of native individuals. The instrument captures the idea that districts with a large initial share of immigrants (as of 1990) are likely to attract a larger share of *new* immigrants coming from the same country of origin because of network effects. So different migration rates from different countries, driven by changing conditions in sending country or policy variations or changes in immigration costs (factors likely to be unrelated to local demand shocks,) have differential impacts on French districts depending on the distribution of these migrants. A potential concern for this type of instrument may arise if the 1990 distribution of immigrants across districts, used to construct the IV, is correlated with economic conditions that affect subsequent changes of productivity and employment in firms. To mitigate these concerns, in the next section we provide several validity test of our instrumental variable, showing that it is not correlated with pre-1996 economic outcomes and trends.

4.1.1. Tests of instrument validity

Past income and productivity shocks, specific to districts, may be persistent over time. If these shocks affected the stock of immigrants in 1990 and are correlated to the change in productivity after 1996 then the exclusion restriction is violated. The instrument may be spuriously correlated to firm's outcomes in the 1996–2005 period. To reduce our concerns about this type of violation of the exclusion restriction we calculate the correlation across districts between the instrument change in the period 1996–2005,³⁸ and the pre-period average economic outcomes in the districts (TFP, domestic market share, capital and trade in year 1996). Table 4 panel A shows no significant correlation between the pre-1996 trend in the instrument and the preexisting average economic outcomes. While this is a reasonable test, it does not rule out correlation of the instrument with pre-1996 trends in the outcomes. A more accurate test would require to regress the trend in our outcome variables before-1996 on the after-1996 trend of our instrument. While we do not have information on French firms' TFP before 1996 we perform two additional tests. First, we test the validity of our instruments, constructed for the second half of the period only (2000–2005) relative to trends of the first part of the period (1996–1999). A strong autocorrelation between earlier demand shock and later changes in the predicted share of immigrants over time would result in significant correlation of pre and post trend and would cast doubt on the validity of the shift-share method. Results reported in Table 4 panel B show that there is no statistically significant correlation between the pre-2000 trend in local outcomes and the post-2000 trend in the instrument.³⁹ As an alternative, instead of using trends in the outcome variables at the firm level, we check the correlation of instruments with pre-existing labor market trends. These data are available at the district level for the years 1982 and 1990 (LFS data). Results reported in panel C of Table 4 show that the instrument is not correlated with preexisting trend in local labor market (district) variables.

³⁷ We could not use directly LFS data to obtain the number of immigrants in France by origin country and year, because being based on micro-census updates, they are less precise as they do not reflect the total population of immigrant in a given year. However, survey data can be safely used to approximate the *share* of immigrants by origin in each year (first term in Eq. (8)).

³⁸ Computed as $\frac{\widetilde{IMMI}_{d,2005} - \widetilde{IMMI}_{d,1996}}{s_{d,2005}^{IMMI} - s_{d,1996}^{IMMI}}$.

³⁹ The only significant correlation is for domestic market share but only at 10% and with the opposite sign that an endogeneity argument would suggest. This is an accurate test for the exogeneity of the instrument after 2000. So, in online appendix Table A6 we show TFP baseline results for the sub-period 2000–2005. Results are qualitatively identical to the baseline estimations on the full sample.

Table 4

Correlation between the IV and economic outcomes in French districts.

Panel A: firms' outcome in the 1996	Change in imputed immigrant share 1996–2005
Total factor productivity OP	0.051 (0.095)
Domestic market share	0.497 (0.597)
Capital	0.371 (0.599)
Total export value	1.625 (1.628)
Number of traded varieties	1.168 (0.826)
Panel B: Change in firms outcomes, 1996–1999	Change in imputed immigrant share 2000–2005
Total factor productivity OP	−0.052 (0.035)
Domestic market share	−0.276* (0.156)
Capital	−0.355 (0.335)
Total export value	−1.121 (0.707)
Number of traded varieties	−0.379 (0.349)
Panel C: Change in local labor market characteristics (1982–1990)	Change in imputed immigrant share 1996–2005
Average wage	−1.457 (1.289)
Average wage native workers	−1.306 (1.391)
Total employment	0.023 (0.364)
Total employment natives	−0.370 (0.349)
Total hours worked	0.353 (0.420)
Total hours worked by natives	0.062 (0.433)
Observations	92

Note: Each row represents the result from a univariate regression having as the dependent variable the change of the imputed district share of immigrant, and the change/level in the economic value listed at the top of each panel for the district as explanatory variable. Observations are 92 French Districts. * = significant at 10% level.

4.1.2. Deviation from the exclusion restriction

While the validity tests discussed above provide evidence consistent with the assumption that the instrumental variable is not correlated with pre-1996 trends, there may exist other omitted variables leading to violations of the exclusion restriction. To deal with concern of lingering endogeneity, we also apply the “plausible exogeneity” test proposed by Conley et al. (2012). This test allows for possible deviations from the exclusion restriction (allowing, that is, non-zero correlation between the instrument and the error) and then tests whether the estimate of the coefficient of interest (s_{dt}^{IMMI}) is robust to a range of these possible deviations. Specifically we relax the exclusion restriction of our IV by assuming that the linear correlation between instrument and error in Eq. (6) above can be in the interval $\gamma \in [-2\delta; 2\delta]$.⁴⁰ The prior on how far the correlation is from exclusion-restriction validity ($\gamma = 0$), can be approximated by regressing the outcome measures on the instrumental variable ($\widehat{IMMI}_{d,t}$), the endogenous variable ($s_{d,t}^{IMMI}$) and all the other control variables included in Eq. (6) (excluding the interaction with the initial characteristics of firms).⁴¹ The estimated coefficients on $\widehat{IMMI}_{d,t}$ (i.e. γ), are reported in Table A10. Following Conley et al. (2012), to estimate the “Local-To-Zero” approximation confidence intervals, we assume γ to be normally distributed with zero mean and variance $\delta = \gamma/2$. In Table A10 of the appendix we report the IV estimates and their 95% confidential intervals produced using Conley et al. (2012) procedure for the plausible range of γ . None of the 95% confidence intervals for β_1 resulting from plausible exogeneity regressions contains zero. This implies that the positive and

⁴⁰ See Conley et al. (2012, Section 4).

⁴¹ Here, we are interested in testing the exclusion restriction of the IV and not its interaction with the initial characteristics of the firm.

Table 5

Immigration in the district and firms' immigrant employment.

	(1)	(2)	(3)	(4)	(5)
Panel A. Dependent variable: Dummy=1 for firms going from zero to positive immigrant employment					
Immigrant share in the district	0.110*** (0.026)	0.181*** (0.039)	0.576*** (0.081)	0.467*** (0.088)	0.406*** (0.082)
(Immigrant share in the district) × (initial TFP below median)				0.261*** (0.116)	
(Immigrant share in the district) × (initial employment below median)					0.376*** (0.098)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS
Observations	119,486	119,482	115,492	98,252	112,599
F-stat of first stage			56.33		
Joint F-stat (Kleibergen-Paap F statistic)				27.44	26.43
Panel B. Dependent variable: log of the number of immigrants in the firm					
Immigrant share in the district	−0.474 (0.426)	0.093 (0.090)	0.424* (0.295)	0.088 (0.299)	−0.051 (0.247)
(Immigrant share in the district) × (initial TFP below median)				1.154** (0.542)	
(Immigrant share in the district) × (initial employment below median)					1.446*** (0.423)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS
Observations	75,960	75,960	74,273	64,348	72,672
F-stat of first stage			54.05		
Joint F-stat (Kleibergen-Paap F statistic)				23.46	20.95

Note: Each column in Panel A and Panel B includes estimates for a different regression. In Panel A, the dependent variable is a dummy equal to one if the firm goes from zero to a positive number of immigrant workers. In Panel B, the dependent variable is the logarithm of immigrant employment. Column (1) includes region and sector fixed effects, and firm level controls (i.e. age of firm and foreign-owned dummy). Column (2) includes region, sector and firm fixed effects and firm level control variables. Columns (3)–(5) include region-by-period, sector-by-period and firm fixed effects and firm level controls. The period considered is 1996–2005. The unit of observation is one firm in one year. *, **, *** indicate significance at the 10%, 5% and 1% confidence level. The standard errors are clustered at the district level.

significant effect of immigrants on firm productivity, as revealed by our IV estimations reported below, is strongly robust to plausible deviations from the exclusion restriction.

5. The effects of immigration on firms

Table 5–12 show the estimated parameters on the explanatory variables of interest for specification (6) using the full sample of firms in our dataset and different firm-level outcomes as dependent variable. These tables share a similar structure, but have different outcome variables. We will describe the main features of the tables when commenting on the estimates presented in Tables 5 and 6 concerning immigrant employment and productivity, respectively.

5.1. Firm's hiring of immigrants

Table 5 shows how the increased supply of immigrants in the district translates into local firms hiring them. In particular, in panel A, the dependent variable is a dummy equal to one if the firm increases the number of immigrants from 0 to a positive value during the considered year. This is an interesting outcome because when hiring the first few immigrant workers the positive impact from complementarity with natives is at its strongest. The main effect – reported in row one of the table – captures how the probability of hiring the first immigrant increases as immigrants in the district rise by one percent of employment. Then, the interaction terms in specifications (4) and (5) capture the additional effect for firms with initial productivity below the median (the coefficient reported in the second row) or with initial size below the median (the coefficient reported in the third row). In panel B of Table 5 we also analyze whether change in district-level immigrant supply affected the number of immigrants hired by the average firm (in logarithms) and whether this effect was stronger in firms with initial productivity (or size) below the median.

If firms are homogeneous in their response to an increase in immigrant supply, new immigrants would be hired in proportion to the firm's size. In this case, a null coefficient should be estimated for the interaction effects shown in columns (4) and (5) of panels A and B of Table 5. Alternatively, if smaller and initially less productive firms, whose initial share of immigrants is small or null, are more inclined to start hiring immigrants because of the stronger complementarity effects, then one would observe a positive effect on firms with initially lower productivity or size. Such scenario would imply positive estimates of the interaction coefficient of columns (4) and (5).

The first two columns of Table 5 show the OLS estimates of the coefficient on the immigrant share of employment in the district. The specification in column (1) includes region and sector fixed effects as controls, while specification (2) also includes firm fixed effects. The 2SLS estimates, on which we focus, are presented in column (3), which also includes a more

demanding set of fixed effects (region-by-period and sector-by-period) as well as controls for individual firm characteristics. In this specification we find that a one-percent-employment increase of immigrants in a district translates into a 0.57% higher probability for the average firm to hire its first immigrant worker (panel A); and to a 0.4% increase in the number of immigrants hired by the firm (panel B). Controlling for firm fixed effects and instrumenting the inflow of immigrants produces a larger point estimate of this effect, relative to OLS estimates. This indicates the presence of omitted variable bias in the OLS specification attenuating the effect.

When allowing for heterogeneous effects we find that firms with initial productivity below the median experienced a stronger increase in their probability of hiring the first immigrant by a further 0.26%, relative to an 0.47% baseline (column 4, Panel A), for an increase in immigrant supply of one percent of the employment in the district. Similarly, the percentage growth in immigrant employment for firms with initially low productivity was much larger (+1.15%) than the baseline (not different from 0) in response to growth in immigrant supply. Column (5) shows heterogeneous response by size. We find that firms smaller than the median firm in 1996 had a significantly higher probability of hiring their first immigrant, and a significantly larger increase in the logarithm of immigrant employment in response to a 1% increase in immigrant supply in the district.

5.2. Firms' TFP growth

A main objective of this paper is studying the effect of immigration on the productivity of firms. Such effect can be positive if immigrants bring complementarity in production and/or if they have positive externalities through choice of efficient technologies or specialization. However, the productivity effect can be negative if immigrants create crowding effects due to the presence of fixed or non-adjustable factors. Table 6 shows the estimated effects of an increase in foreign-born population in the district by one percentage point of employment, on firm's total factor productivity (TFP). We compute the TFP of firms assuming the production function is log-linear, and hence productivity can be calculated as a residual from a log regression. We use Olley and Pakes (1996) approach (henceforth, OP), which accounts for the endogeneity in the use of inputs.⁴² A detailed description of the OP approach and of our calculations is in the appendix Section A2.

In column (1) we show the average effect of immigration estimated using OLS and including only region and sector fixed effects and individual firm controls. Then in column (2) we add firm fixed effects to the previous specification. These estimations may suffer from two problems. First, there can be bias from omitted variables. If the correlation between omitted productivity-increasing variables and the inflow of immigrants is positive, the bias of OLS will be positive. Omitted "productivity" variables negatively correlated with immigration would bias the OLS estimates downward instead. As a consequence of potentially contrasting biases it is hard to predict the sign of the overall bias. Second, in those specifications we do not allow for heterogeneous effects of immigrants across firms. Progressively including fixed effects and then using the IV estimation and, importantly, allowing for heterogeneous effects across firms, should reduce those biases. In particular we observe that the inclusion of firm fixed effects (from columns 1 to 2) increases, but only marginally, the size of the estimated coefficient (i.e. the correlation between immigrant supply and firm TFP) from 0.11 to 0.14. Similarly the 2SLS estimate is marginally larger than the OLS estimate (0.17 vs. 0.14).

The 2SLS estimates represent our preferred specification and column (3) shows the average effect, estimated including the full set of fixed effects as reported in Eq. (6). Increased immigration in the district is associated with faster TFP growth in firms: a ten-percent-of-employment increase in immigrants in the district (which is roughly the average growth of immigrants in French manufacturing between 1995 and 2005) corresponds to a 1.7 log point increase in TFP for the average firm in the district (corresponding to an increase of about 1.7%). More interesting are the estimates allowing for heterogeneous effects, shown in columns (4)–(6) of Table 6. They show a stronger effect of the growth in immigrant supply on TFP for firms that had no immigrants initially. This is consistent with a stronger complementarity effects on firms that previously only hired native workers and implies a 2.7% TFP increase for those firm in response to a ten-percent-of-employment growth of immigrant supply. The estimates in column (5) reveal a stronger productivity effect of immigrants for firms with a low initial TFP level (below the median). Those firms experienced a TFP increase of 3.6 log points for each ten-percent-of-employment increase in the immigrant population in the district. Similarly in column (6), we show that firms with an initial employment level below the median experienced a TFP increase of 3.5 log points for each ten-percent-of-employment increase in the immigrant population in the district, while for firms initially above the median size the effect was only 0.5 log points per ten-percent-of-employment increase.⁴³ The effect of increased immigrant supply on firms' TFP is monotonically decreasing with respect to the initial firm productivity even when we use a finer partition than below/above the median. When interacting the effect with dummies corresponding to quartiles of initial productivity, the magnitude of the coefficients decreases from 0.52 to 0.25 to 0.045 to −0.034 going from the bottom to the top productivity quartile (see Table A7

⁴² As a robustness check we also compute the TFP using a CES aggregation of labor between immigrants and natives. By assuming an elasticity of substitution equal to ten – consistent with Ottaviano and Peri (2012) and Peri (2012) – the CES aggregation of labor is the following: $(Natives^{0.9} + (Immigrants * (ImmWage/NativeWage))^{0.9})^{1.1}$. Estimation results using this approach are reported in appendix Table A6 columns (7) and (8). They are not too different from the simpler approach taken in the main text.

⁴³ While in this and subsequent specifications we include one dimension of heterogeneity at a time and emphasize that initial productivity, size and share of immigrants are strongly (and positively) correlated, we also experimented with including two or three at a time. In general the partition according to initial productivity is the most relevant one in identifying different responses of immigrant share on productivity and other outcomes (these results are available upon request).

Table 6

Immigration in the district and firms' TFP.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: TFP calculation using Olley and Pakes method						
Immigrant share in the district	0.110*** (0.039)	0.141*** (0.023)	0.171*** (0.026)	0.131*** (0.026)	0.014 (0.023)	0.055** (0.026)
(Immigrant share in district) × (zero initial immigrant share)				0.141*** (0.041)		
(Immigrant share in district) × (initial TFP below median)					0.360*** (0.058)	
(Immigrant share in district) × (initial employment below median)						0.296*** (0.047)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Observations	104,432	104,432	100,431	98,056	95,032	98,056
F-stat of first stage			58.86			
Joint F-stat (Kleibergen-Paap F statistic)				28.58	27.92	27.29

Note: Each column includes estimates from a different regression. The dependent variable is the firm TFP calculated using the Olley and Pakes method. Column (1) includes region and sector fixed effects and firm level control variables. Column (2) includes region, sector, and firm fixed effects and firm level control variables. Columns (3)–(6) include region-by-period, sector-by-period, firm fixed effects, and firm level control variables described in the text. The period considered is 1996–2005. The unit of observation is one firm in one year. Standard errors are clustered at the district level. *, **, *** indicate significance at the 10%, 5% and 1% confidence level.

Table 7

First stage results, shift-share imputed immigrants as IV.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable:	Immigrant share	Immigrant share	Immigrant share × zero initial immi sh	Immigrant share	Immigrant share × TFP below 50th	Immigrant share	Immigrant share × empl below 50th
Imputed immigrant share in district	0.644*** (0.084)	0.614*** (0.081)	−0.022 (0.014)	0.630*** (0.082)	−0.027 (0.018)	0.638*** (0.083)	−0.030 (0.019)
(Immigrant share in district) × (zero initial immigrant share)		0.099* (0.058)	0.779*** (0.124)				
(Immigrant share in district) × (initial TFP below median)				0.026 (0.022)	0.714*** (0.109)		
(Immigrant share in district) × (initial employment below median)						0.011 (0.019)	0.708*** (0.106)
Observations	10,0431	98,056	98,056	95,032	95,032	98,056	98,056
F-stat of first stage	58.86	30.20	25.10	29.69	26.59	29.45	26.79
Joint F-stat (Kleibergen-Paap F statistic)			28.58		27.92		25.45

Note: Each column represents a regression. The dependent variable is listed at the top of each column. Columns (1)–(7) always include region-by-period, sector-by-period, firm fixed effects and firm level control variables described in the text. Standard errors are clustered at the district level. The period considered is 1996–2005. The unit of observation is one firm in one year. *, **, *** indicate significance at the 10%, 5% and 1% confidence level.

in the online appendix).⁴⁴ The last row of Table 6 shows the Kleibergen-Paap F-statistics that measures the power of the instruments. In the case of two endogenous variables, the immigrant share and its interaction with the initial characteristics dummy, we use two instruments: the (shift-share) imputed immigrant share described above and its interaction with the $I_i(k_i, 1996 \leq k)$ dummy.

In Table 7 we report more detailed results for the first-stage regressions of the 2SLS procedure. Our instrumental variables are always highly significant in explaining the endogenous share of immigrants in the district and its interactions with $I_i(k_i, 1996 \leq k)$. The F-stat and the joint F-stat are always well above 10, implying the instruments are not weak. Clearly the main instrument is a strong predictor of the main variable (share of foreign-born), while the interacted instrument is a strong predictor of the interaction term.⁴⁵ These results combined with the validity tests of Table 4 imply that our instruments are strong and plausibly exogenous to many economic variables.

5.3. Productivity-related outcomes

In this section, we provide corroborating evidence for the positive productivity effect of immigrant supply using firm outcomes which are positively correlated with productivity and are directly observable, rather than obtained as residual (as the TFP). These outcomes are likely subject to smaller measurement error.

⁴⁴ The coefficients are significant only for the first two quartiles.

⁴⁵ To save space we do not report detailed first stage regressions results for the other firms' outcomes (available upon request). However specification (6) does not change across firms' outcomes and first stage might slightly change only because of missing observations.

Table 8

Immigration in the district and firm's domestic market share or exit probability.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Dependent variable: log of domestic market share						
Immigrant share in the district	0.955** (0.385)	0.990*** (0.159)	0.276*** (0.097)	0.065 (0.132)	−0.061 (0.136)	−0.099 (0.117)
(Immigrant share in the district) × (zero initial immigrant share)				0.667*** (0.247)		
(Immigrant share in the district) × (initial TFP below median)					0.821*** (0.248)	
(Immigrant share in the district) × (initial employment below median)						0.881*** (0.206)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Observations	118460	118460	114455	111609	97395	111609
F-stat of first stage			56.08			
Joint F-stat (Kleibergen-Paap F statistic)				27.02	27.43	26.44
Panel B. Dependent variable: Dummy=1 if the firm exits the market						
Immigrant share in the district	−0.201*** (0.019)	−0.310*** (0.043)	−0.673*** (0.076)	−0.675*** (0.075)	−0.650*** (0.097)	−0.631*** (0.072)
(Immigrant share in the district) × (zero initial immigrant share)				0.022 (0.055)		
(Immigrant share in the district) × (initial TFP below median)					−0.056 (0.088)	
(Immigrant share in the district) × (initial employment below median)						−0.074** (0.030)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Observations	154,742	154,742	151,280	146,040	135,530	146,040
F-stat of first stage			57.32			
Joint F-stat (Kleibergen-Paap F statistic)				29.16	25.67	28.29

Note: Each column of Panel A and of Panel B represents a regression. Column (1) includes region and sector fixed effects and firm level control variables. Column (2) includes region, sector, and firm fixed effects and firm level control variables. Columns (3)–(6) include region-by-period, sector-by-period, and firm fixed effects and firm level control variables described in the text. The period considered is 1996–2005. The unit of observation is one firm in one year. Estimations in panel B do not contain firm level controls as we do not have such information when the firm is not observed in DADS (we squared the dataset to compute the exit dummy, filling the missing after exit with “1”). Standard errors are clustered at the district level. *, **, *** indicate significance at the 10%, 5% and 1% confidence level.

The domestic market share of the firm is one of these measures. It has been used in several papers (e.g. [Blonigen et al., 2014](#)) to proxy for productivity, as productivity-driven cost advantages drive a larger share of sales for a firm in the domestic market. Specifically, we calculate the domestic revenue for each firm and we divide it by the total domestic revenue of all French firms. We report results of this estimation in Panel A (the upper part) of [Table 8](#). These are consistent with those from TFP: a ten-percent-of-employment increase in immigrants in the district produces a 2.8 log point increase in domestic market share when we estimate the effects using 2SLS controlling for the most demanding set of fixed effects. When allowing for heterogeneous effects, according to initial size and initial productivity, most of the impact of immigrants accrues to initially smaller and less productive firms. The impact of a ten-percent-of-employment increase in immigrants on market share is a 6.7 log point increase for firms with no immigrants in 1996 and it is about 8 log points for firms with low productivity or small size (below median) in 1996. The impact on firms above the median of productivity and size is not significant (see columns (5) and (6) in [Table 8](#) Panel A). As further robustness checks, reported in the online appendix [Table A3](#), we use three other proxies for firm productivity as dependent variables: (i) value-added per worker (see [Table A3](#) columns 1 and 2), (ii) TFP calculated as in [Levinsohn and Petrin \(2003\)](#) (see [Table A3](#) columns 3 and 4), and (iii) revenues per worker (see [Table A3](#) columns 5 and 6). Albeit with some differences, all three checks confirm a positive and significant average effect of immigrants and stronger effects on firms with initial productivity below the median.

Another consequence of increased productivity and larger market share for a firm could be the increased probability of survival in the market.⁴⁶ The exit of a firm from the market is rather easy to record and, while a rather “coarse” variable of its performance, it is precisely measured. So, if an increased supply of immigrants benefits firms' productivity, this could also reduce their probability of exiting the market. We test this channel in Panel B of [Table 8](#) where the dependent variable is a dummy equal to one if the firm becomes inactive (conditional on being active for at least two years) and remains such for at least two years. While we do not know from our dataset the reason why a firm ceased its activities, we know that the database includes all firms in France and, hence, exit from the dataset implies the end of the firm's economic activity. The results for this outcome is also in line with those on productivity as immigration reduces firm exit. Using the 2SLS estimates of column (3), we find a ten-percent-of-employment increase in immigrant workers in a district corresponds to

⁴⁶ In [Section 2](#) and within the framework of decomposition (5) we highlighted the possibility that immigration flows affect the entry/exit dynamics of firms.

a 6.7 percentage point *decrease* in the exit probability for the average firm. This effect is larger for firms with employment level below the median in 1996 (see column 6), but the additional effect is small and the probability of exiting decreases only by 7.0–7.1 percentage points, rather than by 6.3 points, for this group. Similarly firms with initial low productivity experience only a small and non-significant additional benefit in terms of exit probability from the increase in immigrant supply.

The fact that immigration can affect the exit probability of firms suggests that selective attrition of firms may drive some of the productivity effect estimated previously, if only the more productive firms survive. To check that this is not a first-order concern, we perform the following check. We estimate Eq. (6) on a balanced panel including only firms that exist over the entire period 1996–2005 (no entry/exit). Results for this checks (reported in the online appendix Table A6) show a similar estimate of the productivity effect of immigrants on this fixed group of firms. This confirms our baseline estimation results revealing no significant bias from selective attrition.

5.3.1. Growth in physical capital

The significant increase in productivity at the firm level identified in the previous section is likely to drive other changes in firms' outcomes. As a large part of the immigrant inflows was made of highly skilled workers, their hiring should have promoted complementarity with physical capital, boosting investment and the adoption of new technologies (e.g. [Per et al., 2000](#)). Then, we should also observe growth in the firm's physical capital stock associated with larger inflows of immigrants.⁴⁷ Moreover, firms may adopt different production techniques in response to larger availability of skilled immigrants, and those can be more capital intensive.⁴⁸

Panel A of [Table 9](#) shows the estimates of specifications similar to those in the previous tables, when the dependent variable is the logarithm of capital value, calculated as the total value of the capital stock of the firm (deflated by its price index). The coefficients tell us whether larger immigration shares in the district stimulate higher net investment in the firm. Column (1) shows the OLS estimates with the more limited set of fixed effects and column (2) includes firm fixed effects as controls. Then in column (3) we show a positive and significant effect on the capital of the average firm using 2SLS: a ten-percent-of-employment increase of immigrants in the district increases the capital stock in the average firm by 8%. When firms hire more immigrants, they significantly increase their capital investment and their capital stock. Columns (4)–(6) show that the effect of immigration on capital investment is stronger for those firms with initially no immigrants, low productivity, or small size (below the median) in 1996. The elasticity of response for initially smaller/less productive firms is about three times greater than the response of other firms.

As a robustness check, reported in Panel B of [Table 9](#), we use an alternative (coarser but easier to collect) measure of firms' capital stock: the number of plants that the firm operates. The results are significantly weaker than those for the total stock of capital (as expected for a variable that changes irregularly and rarely over time). Overall, however, the impact on this outcome is broadly in line with the findings described above. The average effect on the number of firms' plants is positive and is essentially driven by those firms starting in 1996 with 0 immigrants (compare the estimates in columns 3 and 4). Overall, these results show that immigration positively affects the total value of capital as well as the number of plants a firm operates. These findings are clearly complementary to what [Lewis \(2011\)](#) finds for the United States, where prevalently low-skilled immigration is associated with the adoption of unskilled-intensive technology and less mechanization. In France, prevalently high-skilled immigration is associated with increased capital intensity and higher productivity.

5.3.2. Growth in exports

A large body of empirical literature (beginning with [Bernard and Bradford Jensen, 1999](#)) has found that more productive firms are those exporting, and the most productive among them serve more than one market. Thus, the productivity gains from immigration shown in the previous sections could be associated to improvements in the export performances of firms in the manufacturing sector. While several papers have analyzed the connection between immigration and exports at the national and regional level (see [Felbermayr et al., 2012](#) for a recent review), few papers have also looked at the firm-level connection between immigrants and exports. [Hiller \(2013\)](#) and [Hatzigeorgiou and Lodefalk \(2016\)](#) show this connection using Danish and Swedish firm-level data respectively. Most studies find that immigrants have a trade-creation effect on regions and firms and they interpret this as the result of reducing fixed costs of trading with the country of origin of immigrants (e.g. [Peri and Requena, 2010](#)). Firms hiring additional immigrants (from different countries) are more likely to enter additional export markets. Here, we take a slightly different perspective on the trade-migration nexus. The positive productivity effect of immigrants described in the previous section, combined with the reduction in the fixed export cost due to the presence of immigrants, is expected to result in a strong positive impact of immigrants on firm exports. This effect should be larger for initially small and less productive firms, as they experience the highest productivity gain from migration.

Estimates of the effect of immigrants on the (logarithm of the) total value of exports are reported in [Table 10](#) Panel A. They show that an increase in immigrant supply in a district boosts the export sales of firms (column 3). In particular, a

⁴⁷ On the other hand, if immigration was simply an unexpected shift in the supply of labor accompanied by slow and costly capital adjustment, then firms might have experienced a decrease in the capital/labor ratio in the short run and small change in capital stock.

⁴⁸ [Lewis \(2011\)](#) finds that in areas of large low-skilled immigration, firms experience slower growth in their capital–labor ratios.

Table 9

Immigration in the district and firm's capital stock or number of plants.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Dependent variable: log of physical capital						
Immigrant share in the district	0.185 (0.325)	1.006*** (0.198)	0.806*** (0.131)	0.587*** (0.162) 0.638** (0.263)	0.435*** (0.166)	0.474*** (0.162)
(Immigrant share in the district) × (zero initial immigrant share)						
(Immigrant share in the district) × (initial TFP below median)					0.808*** (0.198)	
(Immigrant share in the district) × (initial employment below median)						0.742*** (0.239)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Observations	118,775	118,775	114,928	111,928	98,009	111,928
F-stat of first stage			56.52			
Joint F-stat (Kleibergen-Paap F statistic)				27.28	27.48	26.46
Panel B. Dependent variable: log of number of plants						
Immigrant share in the district	0.096 (0.115)	0.071** (0.031)	0.091* (0.048)	−0.001 (0.069) 0.299*** (0.102)	0.039 (0.089)	0.091 (0.093)
(Immigrant share in the district) × (zero initial immigrant share)						
(Immigrant share in the district) × (initial TFP below median)					0.122 (0.128)	
(Immigrant share in the district) × (initial employment below median)						0.017 (0.152)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Observations	119,486	119,486	115,492	112,599	98,252	112,599
F-stat of first stage			56.33			
Joint F-stat (Kleibergen-Paap F statistic)				27.12	27.44	26.43

Note: Each column of Panel A and of Panel B represents a regression. Column (1) includes region and sector fixed effects and firm level control variables. Column (2) includes region, sector, and firm fixed effects and firm level control variables. Columns (3)–(6) include region-by-period, sector-by-period, and firm fixed effects and firm level control variables described in the text. The period considered is 1996–2005. The unit of observation is one firm in one year. Standard errors are clustered at the district level. *, **, *** indicate significance at the 10%, 5% and 1% confidence level.

Table 10

Immigration in the district and firm's exports or number of foreign markets.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Dependent variable: log of total exports						
Immigrant share in the district	1.103* (0.588)	0.661*** (0.204)	1.275*** (0.343)	0.854** (0.387) 1.524** (0.648)	0.082 (0.298)	0.328 (0.378)
(Immigrant share in the district) × (zero initial immigrant share)						
(Immigrant share in the district) × (initial TFP below median)					2.768*** (0.692)	
(Immigrant share in the district) × (initial employment below median)						2.721*** (0.619)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Observations	87,069	87,069	83,697	81,901	75,600	81,901
F-stat of first stage			58.47			
Joint F-stat (Kleibergen-Paap F statistic)				27.14	28.43	27.70
Panel B. Dependent variable: log of number of foreign markets						
Immigrant share in district	0.544* (0.296)	0.269*** (0.061)	0.611*** (0.147)	0.400** (0.177) 0.681** (0.320)	0.135 (0.129)	0.304* (0.175)
(Immigrant share in the district) × (zero initial immigrant share)						
(Immigrant share in the district) × (initial TFP below median)					1.108*** (0.279)	
(Immigrant share in the district) × (initial employment below median)						0.820*** (0.249)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Observations	87,408	87,408	84,009	82,225	75,882	82,225
F-stat of first stage			58.47			
Joint F-stat (Kleibergen-Paap F statistic)				27.14	28.43	27.70

Note: Each column of Panel A and of Panel B represents a regression. Column (1) includes region and sector fixed effects and firm level control variables. Column (2) includes region, sector, and firm fixed effects and firm level control variables. Columns (3)–(6) include region-by-period, sector-by-period, and firm fixed effects and firm level control variables described in the text. The period considered is 1996–2005. The unit of observation is one firm in one year. Standard errors are clustered at the district level. *, **, *** indicate significance at the 10%, 5% and 1% confidence level.

ten-percent-of-employment increase in immigrants in a district increases the export sales of the average firm by 12.7%. This effect is, as those found above, mainly driven by firms with low initial productivity and small size (see columns 5 and 6).

In a standard trade model with heterogeneous firms (Melitz, 2003), a reduction in the fixed cost of exporting favors in particular small and less productive firms. Immigration, by reducing the fixed cost of exporting, will help small and less productive firms to access additional markets and export more. Moreover, as shown in Section 5.2, immigration boosts the productivity of small and less productive firms in particular, helping them to overcome the productivity threshold to start serving new foreign markets. We test this channel by estimating Eq. (6) on the number of export markets served by firms. In Table 10 Panel B we find a strong positive effect of immigrants on the number of foreign markets (see column 3), which is entirely due to those firms with initially low productivity (column 5) and small size (column 6). As an alternative measure for the extensive margin of trade we also use the number of exported varieties (computed as the number of HS-6 digit items exported by each firm); and find that a positive immigrant supply shock has a positive effect on the number of exported varieties, especially for firms with low initial productivity. These results are shown in Table A4 columns (1) and (2) in the online appendix.⁴⁹

5.3.3. Growth in wages and employment

The higher productivity, higher investment in physical capital, and lower exit probability of firms in response to increased immigrant supply should help native workers in those firms. Immigrants may also have a labor-market competition effect on natives, but the productivity channels highlighted above can attenuate and reverse potential negative effects from that. In a framework similar to that of the previous tables, we consider the logarithm of the average wage of native workers in a firm and the logarithm of native workers employed in the firm as two outcomes. We report the estimated coefficients in Table 11, where we display in Panel A the wage effects and in Panel B the employment effects.⁵⁰ The results of Table 11 Panel A reveal two interesting patterns. An increase in the immigrant share in the district seems to have a strong positive effect on the average wage of natives on average. This is consistent with the productivity effect we showed in Section 5.2. A ten-percent-of-employment increase in immigrants is associated with a 5% increase in native wages in the average firm (column 3 of Table 11). However, the effect on native wages is not stronger in firms initially hiring fewer immigrants (column 4). Moreover, although still positive, it is reduced in firms with low initial productivity (+4.2% in column 5 adding the main effect and interaction) or with small initial size (+3.7% in column 6). These results might appear puzzling if one only considers the productivity effects reported in Table 6. However it is likely due to a composition effect. If, in response to the hiring of immigrants, who are mainly high skilled, the highly-educated native workers move towards firms that hired fewer immigrants, this results in an increase in the average skills of natives in firms with initially high productivity and subsequent low hiring of immigrants. This would imply a positive educational selection of natives in firms hiring fewer immigrants and a positive wage change for the average worker of those firms. Table 12 shows that this is exactly the case. We show the estimated effect of migration on the share of natives in white collar (skilled) jobs relative to the total employment of a firm.⁵¹ Consistently with our explanation, firms with initially low TFP experienced only a small increase in the share of native in white-collar jobs; however, firms with high initial productivity experienced larger and significant increases in the share of native workers in white-collar jobs.

These two results emphasize that the mobility of highly-skilled native workers across firms, in response to immigration, may be an important channel of positive spillovers within a district. Firms that do not hire skilled immigrants still experience part of the benefit via an increase in their skilled native workforce. As long as more skilled workers may also have a productivity effect such a response may diffuse beneficial productivity effects to firms that do not hire immigrants.

To test that such “pure spillover effect” exists and how large it may be, we perform an auxiliary regression, reported in Table A11 of the online appendix. We show the effect of immigration on two types of firms, which both have 0 immigrants at the beginning of the period. The first type, however, hires immigrants over the period (switchers), while the second type never hire immigrants. While the first type of firm should experience both a complementarity and possibly a productivity effect from skilled immigrants, the second will only experience the productivity spillover effects. As expected the firms hiring immigrants experience the highest TFP boost, value added increase and domestic share growth. However, firms not hiring immigrants benefit from immigration in terms of TFP and capital accumulation. The estimated effects are between 50% and 80% of those in the firms hiring immigrants revealing substantial externalities.

Table 11 Panel B shows that initially smaller and less-productive firms expand their employment of natives more than other firms. Given the change in skill composition discussed above, the employment growth must take place mainly through inflow of less-skilled native workers. In columns (3) and (4) of Panel B, we find a null effect of immigration on the employment of natives in the average firm. However, this hides a heterogeneous effect: a ten-percent-of-employment increase in immigrants is associated with a 5% decrease in native employment in firms with high initial productivity and to a 2.2% increase in native employment for firms with low initial productivity (column 5). Immigration clearly induces the reallocation

⁴⁹ In line with these and previous results on TFP we also find that an increase of immigrant supply in a district increases the average export per destination (and variety), with a particularly strong effect for those firms with initially zero migrants in production and for firms with low initial productivity. These results are shown in appendix Table A4 columns (3)–(6).

⁵⁰ Wages are measured at the firm level and hence combine changes in wages of individuals and potential changes in the composition of workers at the firm level (so that selection can take place).

⁵¹ We use white-collar jobs as a proxy for skilled workers since we have data on the occupation of workers but not on their education.

Table 11

Immigration in the district and native workers' wage or employment in the firm.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Dependent variable: log of wage of natives						
Immigrant share in district	0.596*** (0.143)	0.702*** (0.120)	0.488*** (0.073)	0.502*** (0.073)	0.561*** (0.093)	0.573*** (0.090)
(Immigrant share in district) × (zero initial immigrant share)				−0.049 (0.070)		
(Immigrant share in district) × (initial TFP below median)					−0.137* (0.076)	
(Immigrant share in district) × (initial employment below median)						−0.198*** (0.076)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Observations	119,486	119,486	115,492	112,599	98,252	112,599
F-stat of first stage			56.33			
Joint F-stat (Kleibergen-Paap F statistic)				27.12	27.44	26.43
Panel B. Dependent variable: log of native workers in the firm						
Immigrant share in district	−0.077 (0.228)	−0.237*** (0.057)	−0.118 (0.123)	−0.201 (0.164)	−0.490** (0.194)	−0.760*** (0.204)
(Immigrant share in district) × (zero initial immigrant share)				0.188 (0.183)		
(Immigrant share in district) × (initial TFP below median)					0.709*** (0.222)	
(Immigrant share in district) × (initial employment below median)						1.417*** (0.227)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Observations	119,486	119,486	115,492	112,599	98,252	112,599
F-stat of first stage			56.33			
Joint F-stat (Kleibergen-Paap F statistic)				27.12	27.44	26.43

Note: Each column of Panel A and of Panel B represents a regression. Column (1) includes region and sector fixed effects and firm level control variables. Column (2) includes region, sector, and firm fixed effects and firm level control variables. Columns (3)–(6) include region-by-period, sector-by-period, and firm fixed effects and firm level control variables described in the text. The period considered is 1996–2005. The unit of observation is one firm in one year. Standard errors are clustered at the district level. *, **, *** indicate significance at the 10%, 5% and 1% confidence level.

Table 12

Immigration in the district and native white collar workers in the firm.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: white collar natives over total employment						
Immigrant share in district	0.233* (0.126)	0.132*** (0.016)	0.108*** (0.023)	0.138*** (0.031)	0.201*** (0.033)	0.198*** (0.035)
(Immigrant share in district) × (zero initial immigrant share)				−0.085** (0.039)		
(Immigrant share in district) × (initial TFP below median)					−0.187*** (0.037)	
(Immigrant share in district) × (initial employment below median)						−0.202*** (0.047)
Method of estimation	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Observations	119,486	119,489	115,492	112,599	98,252	112,599
F-stat of first stage			56.32			
Joint F-stat (Kleibergen-Paap F statistic)				27.12	27.44	26.43

Note: Column (1) includes region and sector fixed effects and firm level control variables. Column (2) includes region, sector, and firm fixed effects and firm level control variables. Columns (3)–(6) include region-by-period, sector-by-period, and firm fixed effects and firm level control variables described in the text. The period considered is 1996–2005. The unit of observation is one firm in one year. Standard errors are clustered at the district level. *, **, *** indicate significance at the 10%, 5% and 1% confidence level.

and selection of workers across existing firms and thus it changes the relative size and skill composition of firms. In column (6) we show the same kind of evidence using initial size instead of productivity to differentiate firms. Firms with low initial productivity and small size are those that experience the most pronounced productivity boost from immigration and net job-creation, increasing both immigrant and native employment. Their native employment growth is mainly constituted by less-educated workers.

Table 13
Immigration and district specific outcomes.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable:	Number of firms	Employment of natives	Total export	Capital	TFP	Domestic mkt share	Per capita value added	Wage gap (immi/natives)
Immigrant share in district	2.314 (2.010)	2.957 (3.020)	10.53** (0.429)	0.885 (0.930)	0.046 (0.098)	1.383** (0.613)	1.085*** (0.388)	−1.160* (0.592)
Method of estimation	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS
Observations	1001	1001	996	996	910	1001	910	992
Sample	All Firms			All Firms				
Aggregation	Sum across firms in the same district			Weighted average across firms in the same district				
First stage coefficients :								
Imputed Immigrant sh.	0.294***	0.294***	0.293***	0.293***	0.289***	0.294***	0.289***	0.289***
F-stat of first stage	121	167	165	165	146	167	146	160

Note: Columns (1)–(8) include year fixed effects. The period considered is 1996–2005. The unit of observation is one district in one year. *, **, *** indicate significance at the 10%, 5% and 1% confidence level.

5.4. District level regressions

All specifications estimated in the previous sections focus on firm-level outcomes. The results suggest a heterogeneous response of firms to a local increase in immigrant supply, with small and low-productivity firms benefiting more from it. Hence, from an aggregate point of view, the effect of immigration may be more conducive to productivity (and size) convergence across firms than to an average productivity growth. Another important effect can be net firm creation in the district (corresponding to the first term of the decomposition in Eq. (5)), which could magnify employment effects at the firm level. In this section, therefore, we analyze aggregate district-level outcomes to see which firm-level heterogeneous effects survive the aggregation. In Table 13 we show 2SLS panel estimations using data at the district-year level with year fixed effects.⁵² Immigration has a positive, but not significant, effect on both net firm creation and total employment in a district (columns 1 and 2). The firm-level analysis suggests that the last effect may take place with a reallocation of workers from larger (less affected) to smaller (more affected) firms.⁵³ Consistently with firm-level estimation, immigration has a positive effect on exports,⁵⁴ average domestic market share and value added – see columns (3), (6) and (7);⁵⁵ and a positive, but not significant, effect on average TFP and aggregate capital stock.⁵⁶ These results are also consistent with the heterogeneous firm-level results as the strongest positive productivity effects were on initially less-productive firms, driving convergence in productivity rather than an increase in average productivity.

Finally, we estimate the effects of an increase in the supply of immigrants in a district on the wage gap between immigrant and native workers (computed as the ratio between average immigrant over native wage in each district). While improving productivity and opportunities may benefit all workers, the increased supply of immigrants can produce a negative wage effect on other immigrants' wages relative to natives. Consistently with this hypothesis we find evidence of a negative effects on relative immigrant wages in column (8).

6. Conclusion

Using French firm-level data, we studied the effect of an increase in immigrant supply on a set of firm-specific outcomes. As firms vary substantially in terms of initial size and productivity, we analyze also if the response to increased immigrants supply is different according to initial productivity or size. Two empirical regularities emerge from the analysis. First, firms in districts with larger increases in immigrant supply had faster productivity, investment, and export growth, lower exit probability, and larger employment growth than comparable firms in districts with lower inflows of immigrants. Following the literature we use an instrument that proxies the supply-driven inflow of immigrants by distributing new inflows proportionally to the location of historical groups. However, as rarely done in previous research, we test that this instrument is not correlated with pre-period economic trends, reducing the worries of omitted variable bias. We also include a rich set of fixed effects to control for unobserved demand shocks. Hence, while we should be cautious, our result support a causal interpretation of the coefficients. Second, we find that firms with low initial productivity and small size are those that experience the largest productivity, capital, and export growth in response to the local increase in immigrant supply.

⁵² We use the same IV described above.

⁵³ As a further robustness check for the local labor market impact of migration, we replicate our estimations using French Labor Force Survey at the region-year level. Results are reported in the online appendix Table A12. Consistently with results reported in Table 13, we find null effect of immigration on both native workers and wage in the region.

⁵⁴ The point estimate cannot directly be compared with firm-level estimation because of a scale effect from aggregation.

⁵⁵ Coefficients on domestic market share and per capita value added are bigger, but not far from those estimated at the firm level (and reported in Tables 8 and A3).

⁵⁶ To avoid compositional effects in aggregation, we computed a within-district weighted average using the inverse of firm size and productivity in 1996 as weights.

As a consequence the increase in immigrant supply generates some convergence in size and productivity level across firms. This may be due to the fact that smaller, less-productive firms were ready to hire skilled immigrants when their supply increased because of the common EU market, in order to cut costs and/or adopt new technologies and improve specialization. We also show that the positive productivity effect may spill over to firms not hiring immigrants, as they increase hiring of highly-skilled natives. This paper opens up the analysis of a very rich set of local outcomes in response to migration and it describes an interesting chain of effects in response to immigrant supply. The extension of firm-level analysis and the possibility of analyzing effects of immigrants on productivity, mobility, and investment opens new exciting avenues of research in this area, going beyond the usually analyzed labor market effects.

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Supplementary material

Supplementary material associated with this paper can be found, in the online version, at [10.1016/j.euroecorev.2017.05.001](https://doi.org/10.1016/j.euroecorev.2017.05.001).

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